

Jersey Number Recognition Project Proposal

Course: COSC 419B/519B

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Team Number: 4

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1. Introduction

The goal of the Jersey Recognition project is to use deep learning to analyse tracklet data from soccer games and extract jersey numbers from them. Finding jersey numbers in sporting events is critical for keeping up-to-date stats on players to track points, play by play performance and match counters.

The difficulty in finding jersey numbers via tracklets comes from the dynamic nature of sports where the players are constantly moving and changing position. A number 37 upright can look like a 7 if the 3 is being covered on the ground. When there is motion in a given frame, there is a distortion in the image that can also affect results. One aiding factor can be the resolution of the recorded tracklet which means higher quality images with more pixels per frame. This consequently makes the deep learning model work longer as it has to map the objects on a greater pixel density. The SoccerNet database has pre-set tracklets so in real world applications, resolution is not a concern in the scope of this project but is in real world applications.

2. Literature Review & Replicated Results

2.1 Related Work

There are many studies and approaches to jersey number recognition. Some look at individual images [1], others use tracklets [2] to analyse a clip at once. The first challenge faced by both methods is the filtering of keyframes that contain legible numbers that can be analyzed [3-4]. An alternative solution to what we used, involves RoI and Spatial Context Aware filtering to crop the actionable detections before further analysis [4]. Those cropped frames then get put into the next component of the pipeline.

Pose and transform detection is the next critical component of the pipeline. Felzenszwalb and Huttenlocher showed people can be represented in an acyclic tree of connected rectangular structures and used statistical variance to generalize the learning process [5]. ImageNet is the go-to dataset for image annotation and object identifying models [7]. Improvements have been made to the ImageNet model where implementation of NoisyStudent takes labeled and unlabelled images and improves accuracy by 2.0% over the current model [8]. Once the numbered region is identified and isolated then numerical recognition is applied.

For handwritten digits, an ANN with backpropagation was used after scaling digits to the same resolution and turning them monochrome [6]. For image text recognition, there are plenty of datasets to choose from and one such number recognition model PARSeq is trained on 11 of these and has very accurate results [9].

In 2023, the winners of the SoccerNet changed the pipeline by setting a threshold for frames sent to the STR and skipping the frames if they are below [10]. In 2025, the winners altered their training dataset by duplicating segments of their dataset and randomizing features of those duplicated segments [11].

2.2 Replication of State-of-the-Art Results

During the replication process of the pipeline proposed by Koshkina we observed that frame-level introduced some label noise. This was explained in Koshkina's paper as each frame within a tracklet gets the tracklet-level ground truth label as they described as pseudo-labeling but this may not be the actual ground truth due to occlusion, blurriness, ect. Koshkina et al. [3]

We found that the pipeline was very slow and took around 10 hours to replicate the results using a NVIDIA GeForce RTX 4050 GPU using the provided test dataset. The accuracy was close to as reported in the paper, showing the pipeline has an overall good accuracy on the soccernet test dataset.

The challenges we encountered when reproducing the results was set up in Google Collab, which we were able to get the first model to run and hence we got the features but shortly after we were having issues with Google Collab not recognizing the dataset and not being able to run the rest of the pipeline. We opted on running it locally on a laptop with a dedicated gpu where the setup also caused various issues with cloning the needed repositories, accessing files from the GoogleDrive links and dependency issues. Another problem with running the pipeline was the time that it took, and the lack of feedback during the runtime. This caused our team to not be able to have a lot of time to dissect the pipeline and see where the issue/ areas of improvements can be added.

3. Proposed Pipeline

3.1 Overview of Proposed Approach

We will base our model off of the initial pipeline proposed by Koshkina et. al.



The main differences will be when we reach the Pose and STR points of the model. First, in our training dataset we can add random variances to the frames so that the model trains on slightly altered data. Then when the number prediction and legibility classifier is reached, we maintain a confidence score across the tracklet. As of now, the models go frame-by-frame and take a score

for each. With an aggregated confidence score, the entire frame also gets brought into consideration for classifying and number recognition. When we reach the cropping segment, in addition we can color filter the remaining frames such that the text is isolated for numerical STR analysis.

3.2 Model Architecture

The first layer is the eligibility classifier which uses a ResNet34 algorithm trained on ImageNet while fine-tuned on the SoccerNet and random variations on cloned sections of the SoccerNet dataset for additional experience. For pose detection/feature extraction, ViTPose will be used on the MS COCO dataset. Next the data will be cropped and run through an image filter such that only the digits remain visible in the image where our STR layer run on Parseq will be used for final evaluation with an alteration where there is a confidence score across the tracklet.

Our hyperparameters will be set to:

- 25 epochs
- a starting learning rate of 0.001
- momentum of 0.85

3.3 Data Preprocessing & Augmentation

Data is pre-sorted into labelled tracklets for the training and test dataset (1427 train, 1211 test). Then a subsection of that training data (20%) is cloned and the clones are altered with distortion and color transforms. The net sum of the data is then run through the ResNet34 eligibility classifier for filtering. After that is filtered then the ViTPose model analyses the poses, followed by the color filtering and cropping for the numerical digits.

4. Proposed Enhancements & Performance Improvements

4.1 Technical details

Our proposed enhancements aim to improve jersey number recognition accuracy while maintaining computational efficiency. The baseline pipeline consists of pose-based torso cropping, a legibility classifier, and Scene Text Recognition (PARSeq) followed by tracklet-level aggregation.

Instead of performing a simple majority voting across frame predictions, we will maintain a confidence score for each frame prediction and aggregate results using confidence-weighting voting. Frames with low confidence scores will be excluded from the final tracklet prediction. This reduces the influence of blurred or partially occluded frames and improves stability across the tracklet.

We will introduce a threshold mechanism before sending frames to the Scene Text Recognition module. Frames classified as illegible or below a confidence threshold will be skipped. This reduces noise in the input to the STR model and decreases unnecessary computation.

Instead of predicting the entire jersey number as a single class, we will model the prediction at the digit level. This approach improves generalization to unseen jersey number combination and

aligns with findings. To improve robustness, we will apply controlled augmentations during training, such as mild rotation, brightness adjustments, and slight gaussian blur. These augmentations stimulate motion and lighting distortions commonly found in sports footage while preserving digit structure.

4.2 Justification

Research indicates that many frames in sports tracklets do not contain clearly visible jersey numbers. Processing all frames equally introduces noise and degrades recognition performance. Recent SoccerNet challenge results demonstrate that filtering low-quality frames and incorporating confidence-based aggregation significantly improve accuracy. Additionally, digit-wise classification has been shown to generalize better when unseen jersey numbers appear in validation or test sets.

5. Experimental Plan & Evaluation Metrics

5.1 Experiment Design

Detail the planned experiments and configurations, including:

- Baseline comparisons
- Training and validation strategies
- Hardware and software setup

To evaluate the effectiveness of our proposed enhancements, we will conduct controlled experiments comparing a baseline pipeline against modified versions incorporating our improvements. The baseline configuration consists of post based torso cropping (ViTPose), a legibility classifier to filter unreadable frames, PARSeq for STR, and simple majority voting across frames for tracklet-level prediction.

We will then introduce each proposed enhancement independently to isolate its impact on performance. These enhancements include confidence-weighted tracklet aggregation, frame filtering based on legibility and prediction confidence threshold, digit-wise classification instead of holistic number prediction, and controlled data augmentation techniques. After evaluating each modification separately, we will test combinations of improvements to determine whether their effects are complementary.

All experiments will be conducted using the official SoccerNet training and validation splits to ensure consistency and fair comparison. Hyperparameters such as learning rate, batch size, and number of training epochs will remain consistent across experiments to isolate the impact of architectural changes. Early stopping based on validation performance will be used to prevent overfitting. Training will be implemented using PyTorch and executed on NVIDIA GPUs available to the team.

5.2 Evaluation Metrics

Specify the performance metrics that will be used, such as:

- Top-1 and Top-5 Accuracy

- Precision, Recall, and F1-score
- Inference Time and Latency

Model performance will be evaluated primarily using Top-1 accuracy at the tracklet level as this directly reflects correct jersey number prediction for each player sequence. In addition, we will report precision, recall, and F1-score to better understand class-level performance and potential imbalance issues. We will also measure frame-level accuracy to assess how individual frame predictions contribute to overall tracklet results.

To evaluate computational efficiency, we will record inference time per tracklet and monitor the percentage of frames filtered by the legibility and confidence thresholds. This will allow us to analyze trade-offs between accuracy and computational cost. All results will be compared between baseline and enhanced configurations to determine whether the proposed improvements provide measurable gains in recognition performance, speed, or resource efficiency.

6. List of References

- [1] Si et al., "Image Recognition Algorithm Based on Artificial Intelligence and Machine Learning," *Mobile Information Systems*, 2022.
- [2] Li et al., "Unsupervised Person Re-Identification by Deep Learning Tracklet Association," *arXiv*, 2018.
- [3] Koshkina & Elder, "A General Framework for Jersey Number Recognition in Sports Video," *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, 2024.
- [4] Balaji et al., "Jersey Number Recognition using Keyframe Identification from Low-Resolution Broadcast Videos," *arXiv*, 2023
- [5] Felzenszwalb & Huttenlocher, "Pictorial Structures for Object Recognition," *International Journal of Computer Vision*, 2005
- [6] Santoshi et al., "Recognition of Handwritten Digit Using Neural Networks," *IEEE*, 2023.
- [7] Russakovsky et al., "ImageNet Large Scale Visual Recognition Challenge," *arXiv*, 2014.
- [8] Xie et al., "Self-Training With Noisy Student Improves ImageNet Classification," *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020.
- [9] Bautista & Atienza, "Scene Text Recognition with Permuted Autoregressive Sequence Models," *arXiv*, 2022.
- [10] Cioppa et al., "SoccerNet 2023 Challenges Results," *arXiv*, 2023.
- [11] Giancola et al., "SoccerNet 2025 Challenges Results," *arXiv*, 2025.

[12] Liao et al., “Real-Time Scene Text Detection with Differentiable Binarization (DBNet++),” arXiv, 2022.

[13] Du et al., “SVTR: Scene Text Recognition with a Single Visual Model,” arXiv, 2022.

[14] Radford et al., “Learning Transferable Visual Models From Natural Language Supervision,” ICML, 2021.

[15] Xu et al., “ViTPose: Simple Vision Transformer Baselines for Human Pose Estimation,” NeurIPS, 2022.

7. Timeline & Milestones

7.1 Milestones

Milestone 1: Add the aggregate tracklet function to the dataset.

Goal March 10th

Milestone 2: Add the Randomization to the training dataset and setup the image filtering of the numbers

Goal March 17th

7.2 Due dates

Tech demo and Oral slides: Due March 30

Project presentation: April 1st

Project final paper: April 9th

7.3 Timeline

The goal is to have weekly meetings every tuesday or thursday depending on availability with the bulk of the changes done by the 17th of March. Then we can fine-tune hyperparameters and focus on micro adjustments to make the model more efficient and accurate. That will also give time for making the final write-up, presentation and any debugging that will be needed.

8. Team Contributions

<https://github.com/superbolt08/jersey-number-pipeline>

Syed Saad Ali: Responsible for legibility filtering implementation, data augmentation pipeline, debugging dataset loading issues.

Samarth Grover: Lead programmer who ensures that the pipeline runs end-to-end. Keeps track of experiment versions and will need to deliver the final runnable pipeline.

Stephen Lennox: Did research. Responsible for the tracklet-level accuracy, frame-level accuracy, precision, and logging experiment results.

Mehdi Touijar: Translated research findings into actionable implementation guidance for the development team. Wrote section 4 and 5 of the project proposal. Conducted literature review

on state-of-the-art jersey number recognition methods, ZZOM, UniBw Munich VIS, and anything relevant that could improve the project. Responsible for proposing testable modifications and improvements ensuring that the improvements are justified academically.

Ivona Nicetin: Will work on implementing the self - consistency filtering / confidence weight training, addition research, running the pipeline.

Appendix